

DHARMA TEJA GURRAM

📍 Bengaluru, India 📞 +91 9182436977 ✉ gurramdharma925@gmail.com
🌐 linkedin.com/in/dharma925 🐙 github.com/dharma925 🌐 dharma925.github.io/portfolio

SUMMARY

Design Verification Engineer with 2+ years of ASIC/SoC experience, passionate about front-end integration, AI-driven automation, and CI/CD infrastructure. Strong problem-solving and diagnostic skills building LLM-powered workflows, deployment infrastructure, and automated quality gates across Design Sync and Perforce.

EDUCATION

Indian Institute of Technology (IIT) Mandi

Mandi, India

Bachelor of Technology, Electrical Engineering (Honors) — CGPA: 8.68/10

2020 – 2024

Coursework: Digital System Design, Digital MOS LSI Design, VLSI Fabrication, Low Power VLSI Design, Data Science, Estimation and Detection

TECHNICAL SKILLS

- **EDA & Verification:** Cadence Xcelium, SimVision, JasperGold (Lint/CDC), vManager, Virtuoso, Makeflow (ARMCC, ARMClang, ARMLink)
- **Programming & Task Automation:** Python, Bash/Shell, Tcl, Linux
- **Version Control & CI/CD:** Perforce (Helix Core), Design Sync, Git, Bitbucket, Jira, Confluence
- **Design & HDL:** SystemVerilog, Verilog, VHDL, Verilog-A
- **AI Tooling & Software:** Claude API, Claude Code, MCP development, Vectara (LLM/RAG), AWS (EC2, S3, RDS, Lambda, Route53), Node.js, React, Electron

PROFESSIONAL EXPERIENCE

Design Verification Engineer | Texas Instruments, Bengaluru, India

Jul 2024 – Present

- Architected and developed an LLM-powered, state-aware agentic automation stack that transforms manual VLSI workflows into autonomous CI/CD pipelines — maximizing off-hours compute utilization and significantly shrinking project timelines.
 - **State-Aware Agentic Workflow Orchestrator** — plug-and-play framework composing LLM agents into EDA workflows; shipped React/Node dashboards for pipeline-health and monitoring.
 - **Autonomous Regression Lifecycle Manager** — hands-off agent handling workspace syncs, local fixes, and debug reports; reclaims idle off-hours compute capacity, saving ~2 person-months per ~1.5-year cycle significantly reviving the Lead engineer's bandwidth.
 - **Spec-Aware Semantic-Temporal Debug Agent** — MCP-based closed-loop AI debug agent utilizing a custom developed semantic query adapter (simvisdbutil/VCD) to isolate relevant signal windows, cutting debug turnaround time by 6x.
 - **Workspace Health Checker** — Design Sync pre-submit gate running sanity checks; embedded LLM agent diagnoses failures and suggests fixes, eliminating post-sync breakages, conflicts and version jumps.
 - **Release-Triggered Quality Gate** — listens for release emails, auto-launches per-IP Lint/CDC checks, monitors LSF job health, and posts violation summaries to Confluence.
 - **ChatOps Workspace Orchestrator** — Webex-integrated automation interface enabling engineers to securely trigger and monitor Linux workspace operations (syncs, LSF builds) via natural language, bypassing manual login frictions for minor tasks.
- Owned DV for critical IP blocks (ADC, OTP, DPWM, Filter) — testcase/checker/assertion development (including DV-plan-driven testcase-skeleton automation) and sanity/regression closure on every RTL release, in Xcelium/vManager stack.
- **ARM Boot & C-Toolchain Framework** — built the SoC's ARM-cortex core C-compilation/boot infrastructure from scratch: a full make-flow and toolchain integration (ARM CC/ARM Link, ELF-to-hex, RAM/ROM loading) for flexible boot/firmware/application test-case composition and a SimVision custom toolbar for relaunch-free recompiles.
- Drove BU-wide adoption of AI-assisted engineering (Claude Code, custom MCP integrations) across design/verification teams.

Digital Design Intern | Texas Instruments, Bengaluru, India

Jan 2023 – Jul 2023

- UCD3138 Verification — implemented interleaved (2/3-phase) peak-current-mode control from system-level requirements and verified it end-to-end in Xcelium/SimVision: Verilog/AMS debug, full pad-to-pad data-path verification, and bring-up to a working SoC condition with the system engineer.
- Synthesized IP blocks (I²C, PMBus, Timer) for area/power/timing reports and netlists; authored an implementation spec from a communication-interface IP analysis.

PROJECTS

P4Guard — Automated Perforce CI Framework & AI Triage Prototype

Python · Perforce Helix Core · Bash · Claude API

- Deployed a Perforce Helix Core environment (P4D/P4 CLI) to build a CI pipeline utilizing custom Python pre/post-submit triggers to enforce RTL check-in policies and block non-compliant changelists.
- Architected a modular CI orchestrator featuring extensible Python wrappers for simulated EDA stages (lint, synthesis, simulation); integrated a custom task scheduler with Slack reporting and the Claude API for automated failure triage.

NeuronEase — Memristor-Based Hardware Accelerator for Neural Networks

Cadence Virtuoso · Verilog-A · TensorFlow

- Final-year project: designed/simulated a memristor-crossbar accelerator (Stanford ReRAM, Cadence Virtuoso, Verilog-A) for analog multiply-accumulate — weights as crossbar conductances, DAC/ADC I/O; validated via a SPICE-level 2×2 matrix-multiplication demo.
- Trained a compact CNN in TensorFlow (3×3, 12-kernel conv + max-pool + dense, 20,410 INT8 params) and mapped it onto a 9×8 memristor crossbar (row/col decoders, DAC/ADC, conv controller); verified against MNIST at ~12 cycles/pixel, and separately fabricated/characterized Cu/TiO₂/Pt memristive cells (VLSI Fabrication Practicum), measuring I-V resistive switching (I_{on}/I_{off} up to 10⁴).

FCNN on FPGA — Fully Connected Neural Network Accelerator

Verilog · Vivado · TensorFlow · OpenCV

- Built an FCNN in Verilog on a Zybo Z7 FPGA (Vivado) from primitives (adders, multipliers, neurons, dense layers), 16-bit fixed point, TensorFlow-trained; extended into an image-processing pipeline streaming FPGA data, convolving on-chip, results to Python (OpenCV/NumPy).